

# ICT Level 2

## Teaching Guide

Level 2 - Class 7

Level: Level 2

Class: Class 7

Duration: June - January (8 months)

Period: 40 minutes

Curated and developed by

**Hoysala T, Computer Science Teacher**

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

CC BY-SA 4.0 - Free to reuse with attribution - 2026

# Table of Contents

|                                                                        |   |
|------------------------------------------------------------------------|---|
| Curriculum Overview .....                                              | 3 |
| Themes Covered .....                                                   | 3 |
| Pedagogical Approach .....                                             | 3 |
| 5E Model Implementation .....                                          | 3 |
| International Pedagogies Used .....                                    | 3 |
| Lesson Plan Structure (40 Minutes) .....                               | 3 |
| Anatomy of a Lesson Plan (v2, August 2026) .....                       | 3 |
| Assessment Framework .....                                             | 4 |
| Assessment Components (Per Class) .....                                | 4 |
| Resources Required .....                                               | 4 |
| Hardware .....                                                         | 4 |
| Software (Free/Open Source) .....                                      | 4 |
| Dormitory Tasks (Zero Lab Access) .....                                | 4 |
| Capstone Projects .....                                                | 5 |
| Project 1: School Network Design .....                                 | 5 |
| Project 2: Community Data Dashboard .....                              | 5 |
| Unplugged Activities Library .....                                     | 5 |
| Activity 1: Network Relay (Ch 1 - Networking) .....                    | 5 |
| Activity 2: Layer Editing Puzzle (Ch 2 - Graphics) .....               | 5 |
| Activity 3: Audio Editing Cards (Ch 3 - Multimedia) .....              | 5 |
| Activity 4: Algorithm Recipe (Ch 5 - Programming) .....                | 6 |
| Activity 5: Digital Footprint Audit (Ch 6 - Digital Citizenship) ..... | 6 |
| Cross-Curricular Integration Map .....                                 | 6 |
| Mathematics Connections .....                                          | 6 |
| Science Connections .....                                              | 6 |
| Language Connections .....                                             | 6 |
| Digital Citizenship Module .....                                       | 7 |
| Class 7: Being Responsible Online .....                                | 7 |

## Curriculum Overview

ICT Level 2 builds on the foundations established in Level 1, designed for Class 7 students in residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for deeper conceptual understanding.

### Themes Covered

| Chapter | Theme                    | Periods | Key Topics                                       |
|---------|--------------------------|---------|--------------------------------------------------|
| 1       | Networking and Internet  | 12      | Network types, LAN/WAN, Internet services, Email |
| 2       | Advanced Graphics        | 12      | Image editing, Layers, Filters, GIMP             |
| 3       | Multimedia               | 12      | Audio/Video editing, Presentation tools          |
| 4       | Data Handling            | 12      | Spreadsheets, Charts, Formulas                   |
| 5       | Programming with Scratch | 12      | Variables, Conditions, Loops                     |
| 6       | Digital Citizenship      | 12      | Online safety, Cyberbullying, Digital footprint  |

## Pedagogical Approach

### 5E Model Implementation

| Phase     | Duration | Activities                                           |
|-----------|----------|------------------------------------------------------|
| Engage    | 5 min    | Real-world connection, regional metaphors, questions |
| Explore   | 10 min   | Hands-on activity, Driver-Navigator pairs, Lab work  |
| Explain   | 10 min   | Live coding, Concept introduction, C-P-A approach    |
| Elaborate | 10 min   | Extended activity, Parson's problems, Application    |
| Evaluate  | 5 min    | Formative check, Exit tickets, Self-assessment       |

### International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives to Visual blocks to Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator:** Pair programming for lab sessions
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

## Lesson Plan Structure (40 Minutes)

| Time      | Phase                                       | Teacher Activity                                    | Student Activity                                 |
|-----------|---------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| 0-5 min   | Engage -                                    | Present real-world scenario using regional metaphor | Discuss, share observations                      |
| 5-15 min  | Retrieval & Hook<br>Explore - Guided        | Guide hands-on exploration in pairs                 | Work in Driver-Navigator pairs, explore software |
| 15-25 min | Discovery<br>Explain -                      | Live coding demonstration, introduce concepts       | Follow along, ask questions, take notes          |
| 25-35 min | Modelling<br>Elaborate -                    | Assign extension activity or Parson's problem       | Apply learning, solve problems independently     |
| 35-40 min | Application<br>Evaluate - Hinge<br>Q & Exit | Conduct formative assessment, assign homework       | Complete exit ticket, receive dormitory task     |

## Anatomy of a Lesson Plan (v2, August 2026)

Every lesson plan PDF now follows this fixed anatomy. Teachers should read top-to-bottom once before class; the plan is print-ready in black-and-white.

- **Header bar** - Lesson number, topic (italic), duration (40 min), and an archetype badge: CONCEPT (5E inquiry), LAB (Rosenshine I-do/We-do/You-do), or DEBUG (PRIMM Predict-Run-Investigate-Modify-Make). Chapters 2-3 are tagged LAB; chapters 5-6 DEBUG.
- **Success Criteria** - 'I can...' statements derived from objectives; used again during Evaluate for self-assessment ticks.
- **Resources Needed** - Placed at the TOP so materials can be gathered before class. Split mentally into Lab Kit vs Library Paper Pack for the 25/25 station rotation.
- **Key Vocabulary** - Four terms with one-line definitions for EAL learners; pre-teach during Engage.
- **Five phases** (5/10/10/10/5) - Each phase shows TEACHER and STUDENTS blocks side-by-side by left rule colour (blue/green).
- **Differentiation** - Four tiers: Approaching / On Level / Advanced / SEND.
- **Formative Assessment** - Method plus the Computational Thinking skill assessed (maps to the CT rubric in assessments).
- **Hinge Question** - Embedded in Evaluate; if fewer than 80% answer correctly, reteach 3 minutes before Elaborate.
- **Dormitory Task** - Interleaved retrieval: 3 facts from today, 2 connections to previous chapter, 1 question, plus one focused task and a weekly spiral recall.
- **Colophon** - Author, license (CC BY-SA 4.0), version.

## Assessment Framework

### Assessment Components (Per Class)

| Assessment Type          | Count     | Marks Each | Total      |
|--------------------------|-----------|------------|------------|
| Formative (FA-1 to FA-4) | 4         | 30         | 120        |
| Unit Tests               | 4         | 10         | 40         |
| Summative (SA-1, SA-2)   | 2         | 40         | 80         |
| Practical Lab Exams      | 2         | 20         | 40         |
| <b>Grand Total</b>       | <b>12</b> |            | <b>280</b> |

## Resources Required

### Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)
- Projector for demonstrations
- Speakers and headphones

### Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Inkscape
- Audio: Audacity
- Video: Kdenlive
- Programming: Scratch
- Browser: Firefox / Chromium

## Dormitory Tasks (Zero Lab Access)

Lesson plans now embed interleaved 3-2-1 retrieval as the standard dormitory task format. For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Network Diagrams:** Draw and label network topologies on paper
- **Spreadsheet Planning:** Design spreadsheet layouts with formulas on paper
- **Binary Conversion:** Practice decimal to binary conversions
- **Cyberbullying Scenarios:** Discuss and write responses to online safety scenarios
- **Digital Footprint Journal:** Reflect on personal online activities

## Capstone Projects

### Project 1: School Network Design

**Integration:** Science + ICT

**Description:** Students design a network layout for a hypothetical school campus, identifying devices, connections, and resource sharing requirements. They create network diagrams and present their proposals.

**Skills:** Network topologies, LAN/WAN concepts, diagramming, presentation

### Project 2: Community Data Dashboard

**Integration:** Mathematics + ICT

**Description:** Students collect data about their community (population, rainfall, agricultural produce) and create a spreadsheet with charts and formulas to visualize and analyze the information.

**Skills:** Spreadsheet formulas, chart creation, data analysis, digital presentation

## Unplugged Activities Library

### Activity 1: Network Relay (Ch 1 - Networking)

**Duration:** 20 minutes | **Materials:** Paper, tape

**Steps:**

1. Students form a “network” - some are computers, some are routers, some are servers
2. Pass messages (paper slips) from one “computer” to another through “routers”
3. If a “router” is busy (student sits down), messages must find another path
4. Discuss: What happens when parts of a network fail?

**CT Skill:** Network thinking, redundancy, decomposition

### Activity 2: Layer Editing Puzzle (Ch 2 - Graphics)

**Duration:** 15 minutes | **Materials:** Tracing paper, colored pencils

**Steps:**

1. Draw a simple landscape on one sheet (background)
2. On tracing paper, draw only the tree (layer 1)
3. On another tracing paper, draw only the house (layer 2)
4. Stack them. Discuss: Why do graphic artists use layers?

**CT Skill:** Abstraction, decomposition

### Activity 3: Audio Editing Cards (Ch 3 - Multimedia)

**Duration:** 15 minutes | **Materials:** Printed story cards

**Steps:**

1. Give groups a story told in 10 cards

2. Task: Remove 2 cards that are not essential to the story
3. Task: Reorder 3 cards to make the story more interesting
4. Discuss: How is this like editing audio/video?

**CT Skill:** Abstraction, pattern recognition

## Activity 4: Algorithm Recipe (Ch 5 - Programming)

**Duration:** 15 minutes | **Materials:** Paper, pencil

### Steps:

1. Write a recipe for making chai - but use only these words: START, STOP, REPEAT, IF, THEN
2. The recipe must include: boiling water, adding tea leaves, adding milk, checking taste
3. Trade with another group and follow their recipe exactly
4. Discuss: What happens if steps are missing or unclear?

**CT Skill:** Algorithm design, debugging

## Activity 5: Digital Footprint Audit (Ch 6 - Digital Citizenship)

**Duration:** 20 minutes | **Materials:** Paper, pencil

### Steps:

1. List every app, website, and social media you use
2. For each, write: What information does it have about you?
3. Draw a “footprint” showing all your digital traces
4. Discuss: Who can see all this information?

**CT Skill:** Abstraction, evaluation

# Cross-Curricular Integration Map

## Mathematics Connections

- **Ch 1:** Network protocols use binary numbers and addressing
- **Ch 2:** Color theory involves ratios and percentages
- **Ch 3:** Audio/video editing uses timelines (number line concept)
- **Ch 4:** Data handling uses statistics (mean, median, mode, range)
- **Ch 5:** Programming variables are algebraic expressions
- **Ch 6:** Digital ethics involves logical reasoning (IF-THEN)

## Science Connections

- **Ch 1:** Electromagnetic waves carry network signals (Physics)
- **Ch 2:** Light and color perception (Biology/Physics)
- **Ch 3:** Sound waves and human hearing (Physics/Biology)
- **Ch 4:** Data collection follows scientific method
- **Ch 5:** Algorithms model chemical reactions and biological processes
- **Ch 6:** E-waste and environmental science

## Language Connections

- **Ch 1:** Technical writing: network documentation
- **Ch 2:** Visual literacy: interpreting infographics
- **Ch 3:** Script writing for presentations
- **Ch 4:** Data description and report writing
- **Ch 5:** Code comments and documentation
- **Ch 6:** Persuasive writing on digital ethics topics

# Digital Citizenship Module

## Class 7: Being Responsible Online

**Duration:** 3-4 lessons (integrated throughout the year)

### Lesson: Password Security

- Create passphrases: “MyHostelHas20Rooms!” is better than “password123”
- Practice creating strong passwords for different accounts
- Rule: Different accounts need different passwords

### Lesson: Identifying Misinformation

- Analyze 3 WhatsApp forwards: which are real, which are fake?
- Check: Source, date, multiple sources confirming, reverse image search
- Rule: “Verify before you share”

### Lesson: Cyberbullying Response Plan

- Create a 3-step response: Ignore -> Block -> Report to adult
- Practice role-playing scenarios
- Create a class “Safety Pledge” poster

### Lesson: Digital Footprint Management

- Google yourself or your school. What information is public?
- Think before you post: Would you show this to your grandparents?
- Rule: “Your digital footprint is your permanent record”